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Virtual Gold (Hybrid Architecture) x CMU Capstone: Midterm Presentation

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Attachments Virtual Gold (Hybrid Architecture) x CMU Capstone: Midterm Pre...

Meeting records Transcript Recording

Summary

Project scope review and architecture discussion via benchmark evaluation and model calibration

Project Scope and Architecture
Team outlined deliverables and sensitivity data classification requirements for local privacy service. Architecture principles and 18-step request workflow were detailed.

Evaluation and Benchmarks
Evaluation metrics and benchmark results comparing cloud and local models were presented. Local models outperformed baseline and performed comparably to cloud alternatives.

Dataset Trust and Calibration

Discussion focused on establishing ground truth and trust in LLM-generated benchmark data. Model calibration methods and alternative pre-labeled datasets were explored.

Decisions

Aligned

- **Adoption of external benchmark dataset** The participants aligned on sourcing an already-labeled dataset from the internet or Kaggle to establish ground truth.
- **Restriction to a single model configuration** The team agreed to stick with a single model for the current service phase before implementing model swappability.

Next steps

- ☐ [Mark Chu] Send Benchmark Results: Send the preliminary benchmark results and slides to the client for their review. Access these materials from the project drive.
- ☐ [The group] Test Complex Obfuscation: Conduct testing on the removal or obfuscation of complex confidential ideas within the email dataset. Verify the effectiveness of the model using specific system prompts for these scenarios.
- ☐ [The group] Establish trust: Define methods to validate and establish trust in the generated evaluation data set.
- ☐ [The group] Find dataset: Research and acquire a labeled data set from the internet such as Kaggle to serve as a ground truth baseline.
- ☐ [Mark Chu] Explore Gemma: Evaluate the performance of the Gemma 4 Large Language Model within the existing benchmarking framework.
- ☐ [Mark Chu] Create report: Compile a 1-page benchmark report summarizing testing results for all models including Gemma.
- ☐ [Alex] Review results: Examine the provided slides containing initial testing results from the implementation phase.

- [ ] [Alex] Share version: Provide the preliminary version of the service to the team for comparative testing.

Details

- **Project Introduction and Scope**: Mark Chu, Stacy Rosenberg, and Alex discussed the project scope, objectives, and team roles for the local privacy service created by the CMU capstone team for Virtual Gold (VG08), noting that Stacy Rosenberg serves as the faculty advisor.
- **Project Deliverables and Timeline**: Mark Chu outlined the deliverables, which include the VG08 integrated service, an API contract, routing rules, a dashboard, and a 100-email benchmark with evaluation results, set for completion by November 13.
- **Sensitivity Data Classification and Requirements**: Yudi Guo presented the sensitivity classification requirements for stripping private data locally using an open-source model, explaining that the system must automatically judge context, provide confidence scores, and measure performance across 100 labeled emails.
- **Detection Workflow and Sensitivity Tiers**: Yudi Guo described the six-step detection workflow combining rules, Presedio, and a local model, detailing how items are categorized into tiers with specific thresholds (such as 0.2 for high-tier IDs and 0.5 for median-tier names) to determine redaction versus retention.
- **Confidence Score Calibration and Recall Baseline**: Yudi Guo explained that models tend to be overconfident, necessitating score calibration using labeled emails, and noted a baseline recall of 0.63 from prior research where small local models missed one in three items.
- **Evaluation Metrics and Feasibility**: Yudi Guo and Karina Yang detailed that evaluation relies on 100 labeled emails measuring leak rate per tier, over-reduction, calibration, and speed, noting that the service combines existing open-source components with new autonomous decision tiers and thresholds.
- **Architecture Design Principles**: Zhexuan Ye introduced the three core architecture design principles: local execution by default,

model-proposes-code-decides behavior, and a fail-safe approach where unclassifiable items are redacted.

- **Architecture Components and Request Workflow**: Zhexuan Ye and Anmol Sahu detailed the architecture and 18-step request workflow, explaining how the gateway validates requests, segmentation splits text by source context, candidate detection uses rules, Presedio, and a local model, and unionization handles duplicates.
- **Context Assessment and Pseudonymization**: Anmol Sahu explained the context assessment stage where a second local model call evaluates privacy in context, followed by score calibration, threshold comparison in code, and pseudonymization replacing redacted spans with local placeholders like phone_1.
- **Restoration and Failure Path Handling**: Anmol Sahu described the restoration workflow (R1-R4) where the gateway passes restore requests for local map lookup, and the failure path (f1-f3) where model failures trigger a conservative fallback using rules and Presedio, flagging the response as degraded for the caller (VG08) to evaluate.
- **Service Engineering and Monitoring Dashboard**: Zhexuan Ye presented the service engineering plan and Grafana dashboard running locally, which displays eight metrics across three categories (service health and latency, processing readiness, and release outcomes) without storing raw text.
- **Evaluation Data Set Composition and Format**: Karina Yang described the evaluation data set containing 100 realistic fictional executive emails across 15 business areas, 9 industries, and 4 executive roles, encompassing 999 labeled items across 18 categories, deliberate test cases, over-masking test emails, and business confidential statements.
- **Project Schedule and Testing Phases**: Mark Chu outlined the project timeline, including a one-week fall break, core component development through late October, dashboard integration and contract freeze in mid-November, code freeze and final evaluation on November 22, and two overlapping testing phases.
- **Benchmark Results and Model Comparison**: Mark Chu discussed preliminary benchmark results comparing a cloud model (Grok) against local models (Llama 3.1 8-billion parameter and Qwen 2.5 7-billion parameter), reporting that local models outperformed the rules-plus-Presedio baseline and performed comparably to Grok.

- **Complex Use Cases and Testing Inquiry**: Alex inquired about testing complex use cases such as obscuring confidential ideas, and Mark Chu responded that while system prompts can be configured for various use cases, complex use cases have not yet been explicitly tested.
- **Workload Manageability and Development Bottlenecks**: Aarvin George asked about workload manageability, and Mark Chu stated that component linking and model/prompt tuning represent the main bottlenecks, but the overall workload remains manageable for the team.
- **Evaluation Data Set Robustness and Confidence Scoring**: Aarvin George questioned whether the evaluation data set and confidence score generation could be bottlenecks, prompting Karina Yang and Mark Chu to explain that the data set covers complex hard cases and that model prompts instruct the model to output confidence scores between zero and one, which are subsequently calibrated.
- **Evaluation Dataset and Ground Truth Calibration**: Aarvin George emphasized that the evaluation dataset serves as the ground truth necessary for the calibration step of the service. Aarvin George noted that while local models like the Llama 8 billion parameter model and Gemma core have comparable recall—missing one in every three checks—the current VG08 service only improved after gathering real-world interaction data. Aarvin George advised the team to consider how they can trust the generated test cases (00:59:31).
- **LLM-Generated Benchmark Data and Trust**: Mark Chu discussed the team's shift from manually labeling 100 emails to potentially generating benchmark data using a large language model. In response, Aarvin George questioned how the team plans to trust the large language model given that it also outputs confidence scores. Karina Yang confirmed that the team will work on addressing this issue (01:01:10).
- **Alternative Approaches for Establishing Ground Truth**: Aarvin George suggested considering a very strong large language model to generate a baseline that local models can strive to reach, while acknowledging that establishing trust in large language models is a meticulous challenge. Mark Chu proposed an alternative solution of finding a pre-labeled dataset from Kaggle containing privately identifiable information and confidence scores. Aarvin George expressed support for Mark Chu's idea (01:02:25).
- **Model Calibration and Exploratory Testing**: Aarvin George inquired about the calibration process, specifically asking whether scaling and regression

constants are used to map raw scores, to which Mark Chu responded that scaling and regression methods are being considered but require further investigation. Sensing potential overthinking, Aarvin George asked if the team had experimented with the local service yet. Mark Chu offered to share their benchmark results (01:03:42).

- **Model Selection and Benchmarking Methodology**: Mark Chu reported using API keys from Groq and OpenRouter to test Llama and Qwen models. Aarvin George recommended exploring Gemma and sticking to a single model for now, suggesting that a helper or middleware could manage model swappability later as the service improves. Aarvin George praised the overall benchmarking methodology and encouraged the group to evaluate their workload and progress (01:05:02).
- **Implementation Phase and Reporting Deliverables**: Alex expressed anticipation for the upcoming implementation phase and mentioned reviewing the testing results slide and comparing them against a preliminary version of the service. Aarvin George requested that Mark Chu compile a one-page benchmark report for Aarvin George and Alex, ensuring it includes results for Gemma and any newly released models (01:06:23).

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